



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BIOT F424
Course Title : Food Biotechnology
Instructor-in-Charge : SHILPI GARG (shilpi@pilani.bits-pilani.ac.in)
Instructor(s) : Pankaj Sharma (pankajsharma@pilani.bits-pilani.ac.in)

1. Course Description: The food sector is very important from the economical point of view, and is one of the industrial areas where Biotechnology has played and still plays a pivotal role. In this course, a general view of the basic aspects (physical-chemical aspects, structure, spoilage), of foods and their processing and preservation techniques will be taught. Further the most relevant aspects for a future biotechnologist in the food industry will be studied: role of microorganisms and enzymes for the production and transformation of foods and most recent advances in foods made of/with genetically modified organisms. In addition, food safety and quality management along with food waste management will be covered. Finally, case studies of successful entrepreneurship ventures in Food Biotechnology will be discussed.

2. Scope and Objective of the Course: The aim of this course is providing the students with basic knowledge about the applications of Biotechnology in the food industry and in food-related sectors. These include the preservation tools used, production of recombinant proteins and additives to use in food technology, transgenic foods, diagnostic system used in food industry including advanced techniques like biosensors and artificial intelligence, and ethics and safety of food biotechnology.

3. Text Books: “Food Processing and Preservation” Sixth Edition, Author; Sivasankar, B. Publisher; Prentice Hall of India Pvt. Ltd., New Delhi, India, 2009

4. Reference Books:

1. Food Processing Handbook. Edited by James G. Brennan. WILEY-VCH Verlag GmbH & Co. KGaA, 2006, Weinheim, Germany.
2. Food biochemistry and food processing. Edited by Y.H. Hui; 2006, Blackwell Publishing Ltd, UK.
3. Food Microbiology, Martin R. Adams and Maurice O. Moss, Third Edition, 2008, The Royal Society of Chemistry, UK.



4. Course Plan:

Lecture No.	Learning Objectives	Lecture Session	Reference (TB Chap)	Learning Outcome
1	Orientation to the Course	Introduction to Food Science and Technology	Ch1 TB	Food Biotechnology and its applications
2-8	Food Spoilage and Detection; Diseases	Food spoilage: physical, chemical and biological agents, Factors influencing the growth and survival of microorganisms in foods, Food borne illness: Infection & intoxication; Detection of micro-organisms/ spoilage in food; Strategies for the control of spoilage agents.	Ch11,12 TB	Factors responsible for food spoilage, detection of spoilage & prevention of food spoilage. Foodborne diseases, their causative agents, symptoms, and control measures.
9-14	Principles of food preservation	Use of temperature (Heat Sterilization, refrigeration); Radiation etc. for food preservation	Ch15-18 TB	Techniques for effective food preservation and maintenance of food quality and safety.
15-19	Functional Food	Nutraceuticals in health and disease management; Probiotics and prebiotics, Protein foods; Single Cell Proteins; Mycoproteins	Material to be provided	Nutraceuticals and SCP as sustainable sources of nutrition and functional food.
20-22	Enzymes in Food Biotechnology	Enzymes in food science: friend and/or enemy; Modification of enzymes involved in food processing	Material to be provided	Beneficial & adverse roles of enzymes in food science. Use of Biotechnology for modifying enzymes to improve food quality, processing, & shelf life.
23-26	Genetically modified food	Bio fortification; CRISPR based gene editing; GMO crops	Material to be provided	Principles, applications, benefits, risks, and regulatory aspects of genetically modified



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				organisms (GMOs) in food biotechnology.
27-30	Fermented food products	Microbial cultures used in food industries (LAB); Production of Cheese, Beer Brewing, Wine Making, Yoghurt	Ch9, 10, 19, 24 TB	Industrial & commercial production of various fermented food products
31-33	Food additives and colors	Food additives, food colors & sweeteners, emulsifiers; antioxidants etc.	Ch7-8 TB	Safety aspects, and regulatory concepts of additives in food products
34-35	Food Quality And Management	Sensory evaluation of food quality; Ethics and safety of food biotechnology; Food Safety Rules and Regulations	Ch27 TB	Factors affecting food quality, assessment methods, and practices for ensuring safe and nutritious food.
36-37	Food waste management	Utilization of wastes for food production, fuel and fertilizers, Integrated processes & strategies in food production	Material to be provided	Sources and impact of food waste, prevention and sustainable management through resource recovery and reuse.
38-40	Entrepreneurship in Food Biotechnology	Case studies of Successful Entrepreneurship ventures in Food Biotechnology	Material to be provided	Identify opportunities for developing, commercializing, and managing innovative food biotechnology products and ventures.

5. Evaluation Scheme:

S. No.	Evaluation Component	Duration	Weightage (%)	Date, time and Venue	Nature of component
1.	Mid Semester Test	90 min	25%	09/10 FN1	CB
2.	Quizzes	Variable	20%	TBA	CB
3.	Assignment/ Practical/ and Case Studies	Variable	20%	TBA	CB/ OB
4.	Comprehensive Exam	3 hrs	35%	13/12 FN	CB/ OB



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8. Grading Policy:

Award of grades would be guided by the histogram of marks and course average. If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student may not be awarded a valid grade and the case will be reported as “NC”.

9. Chamber consultation hour: Will be announced in the class

10. Notices: Announcements regarding the course will be made in the lecture and will be displayed on Nalanda or will be communicated through email.

11. Academic Conduct Policy: It is expected that all students should attend the classes regularly and follow the highest standard of academic practice when participating in any evaluation component. Having a zero-tolerance for academic dishonesty, any case of misconduct, however minor, will be dealt with strictly with necessary actions and the case may be reported to the Dean, Instruction Division, if necessary.

12. Make-up policy: No make-up will be granted without a genuine case. A certificate from authenticated doctor must accompany the makeup application (only prescriptions/ vouchers for medicines will not be sufficient). Prior permission from IC is strongly recommended. No makeup for in class assignments and quizzes.

Instructor-in-charge
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